Supplementary Protocol: Temporal and Evidence-Source Provenance-Aware Multimodal Misinformation Detection

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1 Eligibility and Rating Mapping

Eligible records are English image–text claims about United States events from 2020–2026. Each record needs an identifiable claim time, an archived or recoverable image, an explicit professional rating, and enough metadata to enforce the temporal cutoff. The mapping accepts three groups of ratings: true or correct; false, fake, or miscaptioned; and unproven, unsubstantiated, or no evidence. Mixed, partially true, satire, opinion, and ambiguous ratings are excluded.

2 Temporal and Group Splits

Training uses records from 2020–2023, validation uses 2024, and the final test uses 2025–2026. The same event, near-duplicate claim, or perceptually identical image cannot cross partitions. Preprocessing parameters and decision thresholds are fitted on training or validation data only.

3 Experimental Configuration

Component	Setting
Evidence discovery	ClaimReview, GDELT, and Common Crawl metadata
Text retrieval	BM25 and BGE-M3
Visual retrieval	SigLIP2 and perceptual hashing
Fusion	Reciprocal-rank fusion followed by reranking
Candidate limits	Rerank 100; retain 20 time-valid items
Graph	Heterogeneous temporal-provenance graph
Decision	Sufficiency gate followed by binary veracity head
Calibration	Validation-fitted temperature scaling
Abstention	Validation threshold targeting 80% coverage
Repetitions	Five seeds after model selection
Uncertainty	10,000 paired event-level bootstrap replicates

4 Ablations and Stress Tests

The ablations remove temporal filtering, dependency clustering, temporal graph features, graph aggregation, textual evidence, visual evidence, the sufficiency gate, and evidence-dropout training. The stress tests inject syndicated copies, add post-cutoff documents that reveal the verdict, remove the earliest source, introduce evidence conflicts, swap captions, and reuse historical images.

5 Dataset Release Policy

The release contains code, URL and hash metadata, derived graph annotations, retrieval logs, and split manifests. It does not include copyrighted article text or images without explicit permission. Fact-check verdicts remain inherited labels, and automatically extracted evidence links are treated as silver evidence.

6 Reporting Checklist

The final submission must report the data snapshot, model and tokenizer revisions, prompts, hardware, precision, search budget, seeds, failures, selection rule, calibration data, decision threshold, elapsed time, and raw prediction locations. Every table must be reproducible from preserved result artifacts.